

		x	x
13	D	Heat lost by volume V of water = heat gained by $\frac{V}{4}$ of water $(\rho V)(4200)(60 - \theta) = \left(\rho \frac{V}{4} \right) (4200)(\theta - 20)$ $\theta = 52^{\circ}\text{C}$	
14	C	At time $t = 0$ the mass is at the amplitude position and the velocity is zero	